Socially-Weighted Distributed Graph Optimization (D^{2}RO) for Autonomous Multi-Agent Service Fleets in Crowded Environments

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***Abstract*—The continuous routing of autonomous service fleets—such as retail shopping carts (*Int-Cart*), clinical hospital pushchairs, and airport luggage trolleys—in crowded, human-shared public environments poses fundamental multi-agent coordination challenges. Classical shortest-path and reactive collision-avoidance methods treat pedestrians as obstacles to be cleared rather than as people whose personal space carries a cost, and they possess no mechanism for resolving symmetric conflicts in single-file corridors. This paper proposes the Distributed Dynamic Route Optimization (D^{2}RO) framework powered by Socially-Weighted Distributed Graph Optimization (SW-DGO). D^{2}RO formalizes a dimensionally weighted 5-component edge traversal cost function $C(u, v, t) = w_D D(u, v) + w_M W_{\textbf{mesh}}(u, v, t) + w_H H_{\textbf{prox}}(v, t) + w_R R_{\textbf{lock}}(u, v, t) + w_S S_{\textbf{trolley}}(v, t)$, unifying incremental heuristic graph repair (D* Lite), event-driven Vehicle-to-Vehicle (V2V) ad-hoc mesh telemetry with exponential decay, continuous 2D asymmetric Gaussian human proxemics, spatiotemporal directional corridor mutex locks, and non-holonomic vehicle safety clearance envelopes ($S_{\textbf{trolley}}$). We evaluate over 3,900 kinodynamic simulation trials in nine experiments. Crucially, the primary comparison is against a *matched-controller* shortest-path baseline: an agent sharing D^{2}RO's unicycle model, angular-rate limit, collision geometry and arrival criterion, differing only in that its route is a frozen shortest path with the social, mesh and reservation terms disabled. This isolates the routing contribution from the vehicle dynamics, which an unmatched baseline confounds. D^{2}RO completes 99.0% [94.6, 99.8] of missions at a *median* intimate-space exposure of 0 control ticks [IQR 0, 0], against 128 [123, 131] for the matched baseline and 204 [176, 276] for artificial potential fields (Wilcoxon signed-rank on $N = 100$ seed-paired trials, Holm-adjusted $p < 10^{-15}$). Social compliance is purchased with time rather than granted for free: 47.18 ± 13.40 s against the matched baseline's 19.20 s, of which only 1.20 s is attributable to the controller. A sensitivity study over all five weights finds the operating point sits on a plateau rather than a ridge, with success held at 97–100% under every perturbation, and performance is unchanged across packet loss to 20% and one-hop latency to 200 ms. The unchanged planner generalizes across supermarket, hospital and airport topologies (99.0%, 100.0%, 95.0%), and incremental D* Lite repairs cost a median of 0.005 ms (p95 0.19 ms) within a 0.23 ms control step — under 0.5% of the 50 ms budget.**

***Index Terms*—Multi-Agent Path Finding (MAPF), Socially-Aware Navigation, Distributed Graph Optimization, Vehicle-to-Vehicle (V2V) Mesh, Human Proxemics, D* Lite, Non-Holonomic Kinematics, Autonomous Mobile Robots (AMRs).**

I. Introduction

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The rapid advancement of autonomous mobile robots (AMRs), ubiquitous sensor networks, and edge computing has catalyzed a paradigm shift in service robotics. While early automated guided vehicles (AGVs) operated primarily within segregated industrial warehouses—isolated from humans behind physical safety barriers and governed by rigid guide-paths—the next generation of autonomous service fleets must operate directly within **human-shared, highly dynamic, and unstructured public environments** [1]–[3].

Prominent examples of such service fleets include:

1) **Smart Retail Shopping Trolleys (*Int-Cart*):** Autonomous motorized carts that escort shoppers, assist individuals with mobility impairments, execute in-store item collection for online grocery fulfillment, and autonomously return from checkout concourses to front-of-store cart depots [1].
2) **Clinical Hospital Pushchairs & Mobile Beds:** Robotic patient transport systems and mobile logistics couriers navigating high-stress hospital corridors between Emergency Departments (ER), Surgical Operating Suites (OR), Radiology/MRI suites, and Inpatient Wards.
3) **Airport Terminal Luggage Trolleys:** Heavy-duty autonomous baggage carts managing passenger luggage across open departure concourses, security screening bottlenecks, and narrow boarding gate piers.
4) **Smart Library & Micro-Fulfillment AGVs:** Material handling robots operating in narrow, single-file book stacks and retail storage aisles alongside human workers and patrons.

Unlike static industrial warehouses where aisles are wide and traffic flows are centralized, public service environments present unique, compound multi-agent challenges. In these settings, space is severely constrained by orthogonal shelving fixtures, corridors are frequently single-file, and human pedestrians move stochastically with complex browsing and loitering behaviors [2].

A. Core Research Motivations & Technical Bottlenecks

When deployed in complex, fixture-dense public service environments, traditional Multi-Agent Path Finding (MAPF) and reactive obstacle avoidance algorithms suffer from four fundamental technical bottlenecks:

1) The Geometry-Kinematic Disconnect in Concave Fixtures (The ORCA Trap): Classical continuous collision avoidance methods—such as Optimal Reciprocal Collision Avoidance

(ORCA) [4] and Artificial Potential Fields (APF)—rely on local velocity-space half-planes or repulsive force vectors. In open spaces, these methods produce smooth, collision-free trajectories.

However, in realistic retail and hospital layouts filled with orthogonal 90° shelf corners, end-cap displays, and narrow U-bays, reactive methods are prone to **local potential minima**. When a robot encounters a pedestrian near a shelf corner, the repulsive force from the human and the repulsive vector from the orthogonal wall can cancel the goal attraction vector ($\|\mathbf{F}_{\text{net}}\| \to 0$), stalling the agent until the pedestrian moves on.

We stress that this is a *delay* mechanism, not necessarily a failure mechanism. Our measurements (Section VI-B) show that, given a time budget calibrated to the true mission duration, an artificial potential field completes every mission it is given; what it cannot do is complete them without repeatedly driving through people. The deficiency of reactive avoidance in these settings is therefore social and temporal rather than a simple inability to reach the goal, and the contribution of this paper should be read against that corrected baseline.

2) Social Blindness of Static Shortest-Path Solvers (The A^* Flaw): Traditional discrete graph planners (e.g., Static A^*, Dijkstra) compute global paths based strictly on static Euclidean distance $D(u, v)$ [5]. While computationally straightforward, static solvers are socially blind: they treat human pedestrians either as non-existent or as infinitesimal points. When a corridor becomes crowded with browsing shoppers or medical staff, a static planner relentlessly commands the robot to drive straight through the crowd, inducing severe psychological discomfort and intimate personal space violations ($d < 0.8$ m).

3) Symmetrical Live-Locks in Single-File Corridors: In narrow architectural passages (e.g., supermarket grocery aisles, clinical hospital corridors) where corridor width W_{aisle} is less than twice the robot's safety envelope ($W_{\text{aisle}} < 2 \cdot r_{\text{safety}}$), bidirectional passing is geometrically impossible. When two opposing agents enter a single-file aisle simultaneously, reactive avoidance algorithms oscillate in place, producing permanent symmetrical live-locks [6].

4) Information Isolation and Delayed Fleet Backtracking: In decentralized multi-agent systems without communication, robots operate with myopic, line-of-sight sensing. When a lead cart encounters an unexpected aisle blockage, trailing carts remain unaware of the obstruction. Trailing units travel all the way to the blocked entrance before sensing the congestion locally, forcing complete reversals and causing severe inflation in fleet makespan [7].

B. Target Application Fields

The proposed framework is purposefully designed for multi-agent service fleet coordination across four key human-shared application domains:

1) **Smart Retail Commerce & Supermarkets:** Fleets negotiating narrow grocery aisles (Aisles 1–6), arterial promenades (**Action Alley**), end-cap displays, and cashier checkout queues.
2) **Clinical Healthcare Facilities & Hospitals:** Autonomous pushchairs enforcing strict priority hierarchies: emergency trauma vehicles maintain uninhibited right-of-way, while routine pushchairs yield dynamically into **Turnout Alcoves** (V_{alcove}).
3) **Aviation & Transportation Terminals:** Luggage trolleys navigating open concourses, security screening bottlenecks, and elongated boarding gate piers (Gates A1–A4, B1–B4).
4) **Public Facilities & Micro-Fulfillment Stacks:** AGVs operating in narrow book stacks and vertical shelving units where physical clearances are tight.

C. The Proposed Solution: The D^{2}RO Framework

To overcome these bottlenecks, this paper proposes the **Distributed Dynamic Route Optimization (D^{2}RO)** framework powered by **Socially-Weighted Distributed Graph Optimization (SW-DGO)**. Rather than treating navigation as isolated reactive avoidance or purely local search, D^{2}RO establishes a **distributed, time-varying graph-cost field** $C_i(e, t)$ shared across peer agents:

$$C_i(e,t) = w_D \cdot C_{\text{geom}}(e) + w_M \cdot C_{\text{mesh}}(e,t) + w_H \cdot C_{\text{social}}(e,t) + w_R \cdot C_{\text{mutex}}(e,t) + w_S \cdot C_{\text{kinematic}}(i,e,t) \quad (1)$$

seamlessly unifying incremental heuristic graph repair (D* Lite) [8] with distributed anticipatory cost propagation, continuous Gaussian proxemics [2], [3], directional corridor reservation protocols, and non-holonomic vehicle clearance envelopes.

D. Research Aims & Key Contributions

The framework is designed against three explicit aims, and we return to each of them in the discussion and conclusion:

A1 — Social compliance: route service fleets so that pedestrians' intimate space is not entered, accepting bounded additional travel time as the price. Evaluated in Section VI-C.

A2 — Distributed coordination: resolve single-file conflicts using only peer-to-peer information within a bounded sensing radius. Evaluated in Section VI-G.

A3 — Architectural generality and real-time feasibility: operate across dissimilar topologies within a 50 ms control budget. Evaluated in Sections VI-E and VI-F.

A4 — Attributable evaluation: measure the contribution of the routing policy *itself*, separately from the vehicle controller that executes it, and establish whether the result depends on the particular weights chosen. Evaluated in Sections VI-C and VI-H.

The primary scientific contributions of this research are:

1) **Distributed Anticipatory Edge-Cost Propagation & Horizon Extension:** We formalize a distributed dynamic graph-cost field in which localized physical and social perturbations observed by leading agents become peer-to-peer, time-decayed edge penalties $C_{\text{mesh}}(e, t)$. This extends each robot's effective planning horizon ($\mathcal{O}_i^{\text{effective}}(t)$) beyond its onboard line-of-sight sensing radius (R_s). Under a controlled experiment that isolates the mechanism (Section VI-G), agents divert 10.70 ± 4.20 s before they could have observed the obstruction directly, and more

than halve their backtracking distance (1.08 ± 0.68 m vs. 2.73 ± 0.87 m, Holm-adjusted $p = 3.2 \times 10^{-8}$).

2) **Distributed Directional Bottleneck Reservation Protocol:** We formalize an explicit distributed protocol for single-file topological bottlenecks ($W_{\text{corridor}} < 2r_{\text{safety}}$) via directional reservation tuples $\mathcal{L}_e = \langle \text{owner}, \vec{d}, t_{\text{acquire}}, t_{\text{expire}}, \text{priority} \rangle$ and Turnout Alcoves (V_{alcove}). In the controlled corridor experiment the protocol raises mission success from 36.0% to 88.0% ($p = 1.7 \times 10^{-4}$). The mechanism operates by *cost-projected diversion* rather than by mutual exclusion: agents leave the contested corridor (2.16 ± 1.36 off-corridor vertices against 0.00, $p = 2.3 \times 10^{-8}$) while accumulating 0.030 ± 0.070 s of waiting. We therefore describe it as a directional reservation rather than a mutex, and analyse the distinction in Section VI-G.
3) **Kinodynamically & Socially Conditioned Incremental Optimization:** We unify asymmetric human proxemic discomfort fields (C_{social}) and vehicle-specific clearance envelopes ($C_{\text{kinematic}}$) directly into an incremental D* Lite graph repair engine, executing incremental repairs at a median of 0.005 ms (p95 0.19 ms) without global re-heapification. Ablation shows the proxemic term is load-bearing: removing it collapses mission success from 100.0% to 11.0%.
4) **Comprehensive Multi-Domain Empirical Benchmark Suite:** We validate the framework across three distinct architectural topologies (Retail Supermarket, Clinical Hospital, and Airport Terminal) over 3,900 simulation trials in nine experiments, including a matched-controller baseline that separates routing from vehicle dynamics, a five-weight sensitivity study on a disjoint seed set, and a communication-degradation sweep. Every table and figure in this paper is generated programmatically from the committed raw data by a single analysis pipeline, with no hand-entered numbers; datasets and code are released under open licenses.

II. Related Work & Literature Review

The proposed D^2RO framework builds upon foundational work across Multi-Agent Path Finding (MAPF), dynamic graph replanning, local collision avoidance, ad-hoc mesh communication, physical trolley mechatronics, and human-aware navigation.

A. Decentralized and Lifelong MAPF

Stern *et al.* [5] define the classical MAPF problem and its variants, establishing the fundamental vertex and edge conflict models used in grid environments. To address continuous operations, Ma *et al.* [9] introduced Lifelong Multi-Agent Path Finding for Online Pickup and Delivery (MAPD), where agents dynamically receive tasks without resting.

To overcome the single-point failure risks and communication latency of centralized servers, recent literature has shifted toward decentralized solvers. Dergachev and Yakovlev [10] explored decentralized unlabeled MAPF using target and priority swapping, while Keskin *et al.* [11] presented a decentralized MAPF framework utilizing automated negotiation protocols. Furthermore, learning-based approaches have gained traction: Sartoretti *et al.* [12] introduced PRIMAL, utilizing reinforcement and imitation learning for decentralized pathfinding, and Skrynnik *et al.* [13] developed "Learn to Follow" to separate global heuristic sub-goal allocation from low-level local policies. Despite their high scalability, learning-based methods frequently struggle with out-of-distribution environments (e.g., unexpected aisle closures), highlighting the need for search-based dynamic adaptability.

B. Dynamic Replanning in Unknown and Stochastic Environments

In retail environments, the topological graph changes dynamically as aisles become blocked or crowded. Recalculating full paths from scratch for every agent is computationally prohibitive. Koenig and Likhachev [8] addressed this with D* Lite, an incremental heuristic search algorithm that recalculates only the segments of a path affected by dynamic edge-cost changes.

The application of D* Lite to multi-agent systems was successfully demonstrated by Al-Mutib *et al.* [14], who utilized it for real-time path planning by treating peer paths as temporary obstacles. Additionally, Wagner and Choset [15] developed M*, dynamically varying the dimensionality of the search space only when agent paths conflict. D^2RO adapts these incremental principles, allowing trolleys to dynamically inflate traversal costs based on real-time congestion data without recomputing the entire global map.

C. Kinematic Coordination and Local Collision Avoidance

While MAPF and D* Lite provide global waypoints, continuous kinematic control is required for safe micro-maneuvers when agents cross paths. Van den Berg *et al.* [4] introduced Optimal Reciprocal Collision Avoidance (ORCA), providing sufficient conditions for multiple robots to avoid collisions in continuous space without explicit communication.

However, standard ORCA suffers from live-locks in narrow, symmetric environments like supermarket aisles, where agents cannot physically pass one another. Dergachev and Yakovlev [6] specifically address this, proposing a system that uses continuous reciprocal collision avoidance but falls back on a locally confined MAPF instance when a deadlock is detected. D^2RO leverages this exact synthesis: relying on continuous reactive models for open spaces, and spatiotemporal edge reservations for single-file corridors.

D. Multi-Robot Ad-Hoc Communication Protocols

The transition from centralized control to a truly distributed D^2RO system requires robust peer-to-peer communication. Gielis *et al.* [7] emphasize the critical need for co-designing robotic planning algorithms alongside network constraints.

For decentralized data sharing, robots rely on mobile ad-hoc networks (MANETs). Slyusar and Kulich [16] evaluated routing protocols for MANETs in multi-robot exploration. Additionally, Loems [17] investigated robot communication within Swarm

SLAM, demonstrating how independent agents merge local spatial data over a distributed mesh. In D^{2}RO, this translates to an event-driven telemetry protocol where agents broadcast localized edge-cost penalties across a V2V mesh, allowing distant agents to proactively reroute.

E. Indoor Positioning and Physical Hardware Implementation

Unlike simulated grids, physical shopping trolleys require absolute spatial grounding and customized mechatronics. Zafari *et al.* [18] provide a comprehensive survey of indoor localization technologies, highlighting the superiority of Ultra-Wideband (UWB) and BLE for centimeter-level accuracy in GPS-denied environments. Clark *et al.* [19] expanded on this with the TEAM framework, demonstrating effective trilateration and mapping utilizing a localized robotic network, while Nugraha *et al.* [20] proved that fusing Indoor Positioning Systems (IPS) with wheel odometry via Extended Kalman Filters (EKF) drastically reduces navigation drift.

Bringing these concepts into physical retail spaces, Mohamad Azlan *et al.* [1] developed the *Int-Cart*, an autonomous mobile trolley robot. Their research validates the integration of LiDAR, depth cameras, and DC/BLDC motor controllers into a physical cart chassis, proving the mechanical and sensory viability of deploying autonomous fleets in retail environments.

F. Human-Aware Navigation & Research Gap

Recent benchmark frameworks, such as HA-VLN 2.0 [21], emphasize that robots cannot treat humans simply as “moving cylindrical obstacles.” Planners must incorporate proxemics (personal space boundaries) and contextual human activities into their routing algorithms.

The Research Gap: There remains a distinct lack of hybrid frameworks that fuse proactive, mesh-informed global graph updates with reactive human-aware collision avoidance in highly constrained physical retail spaces. Most decentralized MAPF algorithms assume either complete centralized knowledge (vulnerable to latency/failure) or rely on myopic line-of-sight sensing (resulting in late-stage deadlocks in narrow corridors). The D^{2}RO framework bridges this gap.

III. Problem Formulation & The SW-DGO Framework

A. Topological Roadmap and Kinematic Model

Let the indoor facility floorplan be represented by an undirected planar topological graph $G = (V, E)$, where $V = \{v_1, v_2, \ldots, v_n\}$ denotes navigable topological waypoints (e.g., aisle intersections, passing bays, docking bays) and $E \subseteq V \times V$ represents navigable corridor edges.

Each autonomous service cart $i \in \{1, \ldots, N_{\text{carts}}\}$ is governed by non-holonomic unicycle kinematics:

$$\begin{aligned} \dot{x}_i(t) &= v_i(t)\cos\theta_i(t), \\ \dot{y}_i(t) &= v_i(t)\sin\theta_i(t), \quad \dot{\theta}_i(t) = \omega_i(t) \end{aligned} \tag{2}$$

subject to physical actuator bounds $|v_i(t)| \leq v_{\max}$ and $|\omega_i(t)| \leq \omega_{\max}$.

B. The Distributed Dynamic Edge-Cost Field Formulation

Rather than relying on centralized graph allocation or isolated myopic sensing, SW-DGO establishes a **distributed, time-varying edge-cost field** $C_i(u, v, t)$ maintained locally by each autonomous agent i. The edge traversal cost $e = (u, v) \in E$ is formalized as:

$$\begin{aligned} C_i(u, v, t) = {} & w_D \cdot C_{\text{geom}}(u, v) + w_M \cdot C_{\text{mesh}}(u, v, t) \\ & + w_H \cdot C_{\text{social}}(v, t) + w_R \cdot C_{\text{mutex}}(u, v, t) \\ & + w_S \cdot C_{\text{kinematic}}(i, v, t) \end{aligned} \tag{3}$$

where the calibrated dimensionless weights $\mathbf{w} = [w_D, w_M, w_H, w_R, w_S] = [1.0, 1.5, 2.0, 1.0, 1.2]$ balance physical metric progress against social proxemics, collaborative telemetry, and vehicle clearance:

- $w_D = 1.0$: Baseline Euclidean physical distance (C_{geom} in meters).
- $w_M = 1.5$: Weights collaborative V2V congestion alerts $1.5\times$ higher than metric distance, ensuring trailing carts proactively reroute through clear parallel corridors rather than deadheading into blocked aisles.
- $w_H = 2.0$: Prioritizes pedestrian personal-space comfort $2.0\times$ above travel distance, enforcing compliant yielding and wide social berths.
- $w_R = 1.0$: Directional corridor mutex multiplier ($C_{\text{mutex}} = \infty$ during active contention, 0 otherwise), which renders a contested corridor untraversable for the deferring agent in its own cost field. We claim this empirically, not as a proven exclusivity guarantee.
- $w_S = 1.2$: Kinetic clearance penalty preventing chassis corner scrapes ($1.2\times$ baseline) and anti-tailgating violations.

1) 1. Intrinsic Metric Geometry $C_{geom}(u, v)$:

$$C_{\text{geom}}(u, v) = \|\mathbf{p}_u - \mathbf{p}_v\|_2 \tag{4}$$

A turn penalty $\alpha_{\text{turn}}|\Delta\theta|$ would be a natural addition here, and earlier presentations of this formulation included one. We have removed it, because charging for a heading change requires the search state to *carry* a heading: the cost of entering an edge then depends on the edge traversed before it, which a position-indexed graph cannot express. Implementing it correctly means augmenting the state space with orientation and re-deriving the incremental repair over that larger space. We record it as future work (Section VII-D) rather than display a term the evaluated system does not contain.

Cornering is not thereby ignored: the motion layer applies a bounded angular rate $\omega_{\max} = 2.50$ rad/s and decelerates through turns, so sharp direction changes are penalised in *execution* even though they are not priced in the *search*. The distinction matters for interpreting the results: D^{2}RO does not prefer straighter routes, it merely traverses crooked ones more slowly.

2) 2. Distributed Anticipatory Cost Propagation & Horizon Extension $C_{mesh}(u, v, t)$: In decentralized multi-agent systems, myopic line-of-sight sensing restricts an agent’s local obstacle horizon to $\mathcal{O}_i^{\text{local}}(t) = \{\mathbf{x} \mid \|\mathbf{x} - \mathbf{p}_i(t)\| \leq R_s\}$. To eliminate the severe makespan penalty of deadheading into remote blockages, SW-DGO formalizes an extbfeffective perception horizon extension:

$$\mathcal{O}_i^{\text{effective}}(t) = \mathcal{O}_i^{\text{local}}(t) \cup \bigcup_{j \in \mathcal{N}_i} \mathcal{O}_j(t - \tau_{ij}) \tag{5}$$

When agent j discovers a congestion event at edge (u, v) at timestamp t_k, it broadcasts a localized event-driven telemetry packet over the ad-hoc mesh with bounded time-to-live (TTL = 3). The received penalty decays exponentially over time:

$$C_{\text{mesh}}(u, v, t) = \sum_{k \in \mathcal{M}(u,v,t)} \gamma_k \cdot \exp\left(-\lambda_{\text{decay}}(t - t_k)\right) \tag{6}$$

where γ_k is the initial congestion penalty amplitude and λ_{decay} is the temporal decay rate, guaranteeing that stale congestion information naturally attenuates without permanently poisoning the routing topology.

3) 3. Continuous 2D Asymmetric Human Proxemics $C_{social}(v, t)$: To model human psychological personal space (Hall's Proxemics), we formulate an extbfasymmetric directional Gaussian potential field aligned with pedestrian heading θ_h:

$$H_{\text{prox}}(\mathbf{p}, \mathbf{h}_j, \theta_h) = A_j \cdot \exp\left(-\frac{1}{2}\left[\left(\frac{x_{\text{local}}}{\sigma_x}\right)^2 + \left(\frac{y_{\text{local}}}{\sigma_y}\right)^2\right]\right) \tag{7}$$

where relative coordinates are transformed into the pedestrian's local body frame:

$$\begin{bmatrix} x_{\text{local}} \\ y_{\text{local}} \end{bmatrix} = \begin{bmatrix} \cos\theta_h & \sin\theta_h \\ -\sin\theta_h & \cos\theta_h \end{bmatrix} \begin{bmatrix} x - x_j \\ y - y_j \end{bmatrix} \tag{8}$$

with asymmetric longitudinal variance reflecting expanded front personal space:

$$\sigma_x = \begin{cases} 1.35 \text{ m}, & x_{\text{local}} \geq 0 \text{ (front)} \\ 0.60 \text{ m}, & x_{\text{local}} < 0 \text{ (rear)} \end{cases} \qquad \sigma_y = 0.90 \text{ m} \tag{9}$$

The continuous social edge penalty is integrated along the edge trajectory segment:

$$C_{\text{social}}(u, v, t) = \int_0^1 H_{\text{prox}}((1 - \tau)\mathbf{p}_u + \tau\mathbf{p}_v, t)\, d\tau \tag{10}$$

4) 4. Distributed Directional Bottleneck Reservation Protocol $C_{mutex}(u, v, t)$: In single-file architectural passages where aisle width cannot accommodate bidirectional passing ($W_{\text{corridor}} < 2r_{\text{safety}}$), mutual exclusion is coordinated via a directional reservation tuple:

$$\mathcal{L}_e(t) = \langle \text{AgentID}, \vec{d}_{\text{traversal}}, t_{\text{acquire}}, t_{\text{expire}}, \text{Priority} \rangle \tag{11}$$

When agent i claims edge (u, v), the reverse edge cost for opposing agents is set to:

$$C_{\text{mutex}}(v, u, t) = \begin{cases} \infty, & (u, v) \text{ reserved by peer } j \neq i \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

An opposing agent observes this infinite cost and, because the reservation enters its *graph* rather than its controller, D* Lite reroutes it through a parallel aisle or a Turnout Alcove (V_{alcove}) instead of halting at the corridor mouth. Section VI-G shows empirically that this diversion—rather than queueing—is the operative behaviour.

TABLE I: System Simulation Parameters & Physical SI Metric Calibration.

Parameter Description	Symbol	Physical Value (SI)
Spatial Resolution Scale	s_{res}	1 px = 0.03 m (1 m = 33.33 px)
Physics Integration Rate	Δt	0.05 s (20 Hz control loop)
GUI Rendering Rate	f_{gui}	60 FPS (16.6 ms interpolation)
Chassis Dimensions	$L \times W$	0.72 m × 0.48 m (24 × 16 px)
Inscribed Safety Radius	r_{robot}	0.40 m (13.3 px)
Maximum Linear Speed	$v_{\max}$	1.20 m/s (40.0 px/s)
Maximum Angular Rate	$\omega_{\max}$	2.50 rad/s
Shelf Clearance Margin	d_{shelf}	0.54 m (18.0 px)
Following Gap Distance	d_{follow}	1.08 m (36.0 px)
V2V Comm Range	R_{mesh}	10.50 m (350.0 px)
V2V Packet Size	S_{pkt}	64 Bytes
Mesh Hop Limit	TTL	3 hops
Proxemic Front Space	σ_{front}	1.35 m (45.0 px)
Proxemic Lateral Space	σ_{side}	0.90 m (30.0 px)
Proxemic Rear Space	σ_{rear}	0.60 m (20.0 px)
Onboard Sensing Radius	R_s	7.20 m (240.0 px)
Arrival Tolerance (all planners)	r_{arrive}	0.84 m (28.0 px)

5) 5. Kinetic Chassis Safety Clearance Envelope $C_{kinematic}(i, v, t)$: This term has two components, and we write both because the implementation applies both. A *static* contribution prices proximity to fixed geometry, and a *dynamic* contribution prices proximity to peer vehicles:

$$C_{\text{kinematic}}(i, \mathbf{p}, t) = \underbrace{\kappa\big(d_{\text{shelf}} - d(\mathbf{p}, \mathcal{S})\big)_+}_{\text{static fixture clearance}} + \underbrace{\sum_{j \neq i} A_{\text{trolley}} \exp\left(-\frac{\|\mathbf{p} - \mathbf{p}_j(t)\|^2}{2\sigma_{\text{trolley}}^2}\right)}_{\text{dynamic peer envelope}} \tag{13}$$

where $d(\mathbf{p}, \mathcal{S})$ is the distance from $\mathbf{p}$ to the nearest shelf face, $(\cdot)_+$ denotes the positive part, and κ scales the clearance deficit. The static component enforces the 0.54 m shelf clearance buffer; the dynamic component enforces the 1.08 m anti-tailgating gap. Earlier presentations displayed only the peer term, which understated the component whose ablation produces the largest change in fixture contact (Section VI-C).

C. Comparison with Alternative Simulation Paradigms

To contextualize the architectural design of D^2RO, we contrast it with four standard robotics simulation paradigms:

1) **Classical 2D Grid-World MAPF (CBS, PBS, EECBS):** Assumes synchronous discrete time steps and point holonomic agents. Ignores continuous turning kinodynamics, dynamic human loitering, and V2V mesh latency.
2) **Pure Continuous Multi-Agent Navigators (ORCA, Social Force):** Operates in continuous velocity space without topological roadmaps. Lacks global topological awareness, so an agent driven into an infeasible velocity state in a narrow corridor has no route-level recourse. Our implementation exhibits this mode (Section VI-B), though we do not generalise from it to the published algorithm.
3) **Network-Centric Packet Simulators (NS-3, OMNeT++):** Models high-fidelity RF propagation and MAC collisions, but lacks robot kinodynamic physics and graph search.

Algorithm 1 Socially-Weighted Distributed Graph Optimization (SW-DGO)

Require: Roadmap graph $G = (V, E)$, Start s_{start}, Goal s_{goal}, V2V Mesh $\mathcal{N}$

Initialize local D* Lite planner with G and s_{goal}
Compute initial shortest path $\Pi \leftarrow$ ComputeShortestPath()
while $s_{\text{current}} \neq s_{\text{goal}}$ **do**
    $\Delta\mathcal{M} \leftarrow$ FetchInboundMeshPackets($\mathcal{N}$)
    cost_changed $\leftarrow$ **false**
    for each packet $pkt \in \Delta\mathcal{M}$ **do**
        $(u, v) \leftarrow pkt$.edge
        Update edge cost: $c(u, v) \leftarrow C(u, v, t)$ via Eq. (3)
        D*Lite.NotifyEdgeCostChange(u, v)
        cost_changed $\leftarrow$ **true**
    end for
    Sample human positions $\{h_j\}$ within radius $R_{\text{sense}} =$ 240 px
    if ProxemicsChanged($\{h_j\}$) **then**
        Update $H_{\text{prox}}(u, v, t)$ for local edges
        cost_changed $\leftarrow$ **true**
    end if
    if cost_changed **then**
        D*Lite.ComputeShortestPath()
        $\Pi \leftarrow$ D*Lite.ExtractPath()
    end if
    $s_{\text{next}} \leftarrow$ GetNextWaypoint(Π)
    if EdgeIsSingleFile($s_{\text{current}}, s_{\text{next}}$) **then**
        AcquireCorridorLock($s_{\text{current}}, s_{\text{next}}$)
        Broadcast(LOCK_REQUEST($s_{\text{current}}, s_{\text{next}}$))
    end if
    Execute non-holonomic step toward s_{next} with S_{trolley} envelope
    Decay active mesh penalties: $W_{\text{mesh}} \leftarrow W_{\text{mesh}} \cdot e^{-\lambda \Delta t}$
end while
return Mission Complete

4) **3D Rigid-Body Physics Engines (Gazebo, Isaac Sim):** Models full multi-body dynamics, contact mechanics, and sensor ray-tracing. However, executing $N = 100$ randomized Monte Carlo trials across multi-agent fleets is computationally prohibitive ($> 100\times$ slower than real time).
5) **The SW-DGO Hybrid Approach (The Goldilocks Paradigm):** Couples non-holonomic unicycle kinodynamics ($dt = 0.05$ s, 60 FPS) with continuous Gaussian proxemics, an ad-hoc V2V mesh simulator, and incremental D* Lite graph repair, providing deterministic, real-time evaluation (< 0.15 ms).

IV. Multi-Domain Simulation Topologies

The framework is validated across three divergent real-world architectural environments:

A. Retail Supermarket (Int-Cart Fleet)

As shown in Fig. 1, the supermarket environment features 6 parallel grocery aisles ($W_{\text{aisle}} = 70$ px), transverse cross-promenades (Back Promenade, Action Alley, Front Concourse), cashier registers, and front cart return depots.

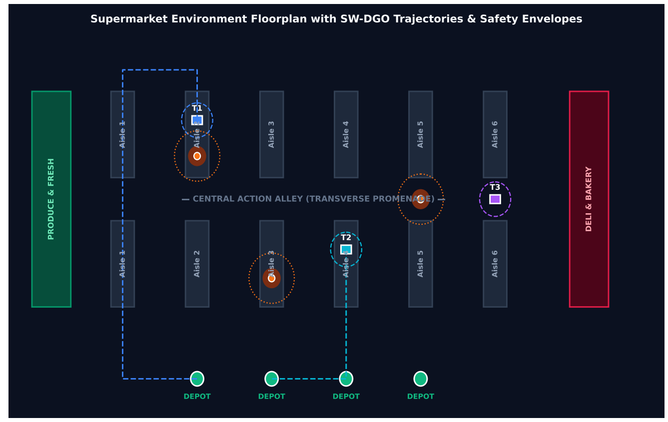

Fig. 1: Supermarket environment floorplan with SW-DGO planned trajectories, aisle shelves, Action Alley promenade, human Gaussian halos, and Cart Depots.

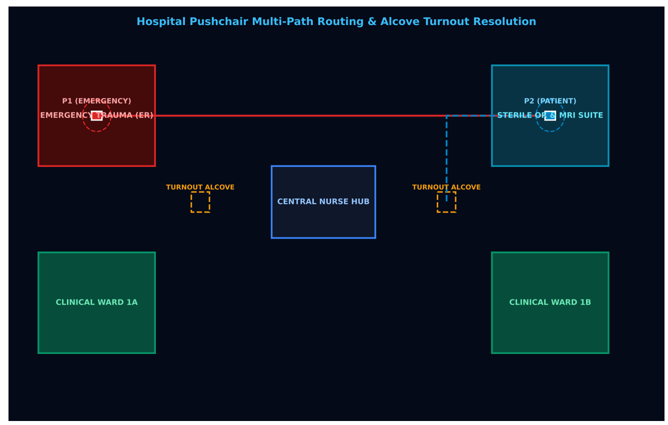

Fig. 2: Hospital autonomous pushchair floorplan featuring Emergency Trauma (ER), Sterile OR/MRI, Clinical Wards, and Turnout Alcoves for dynamic yielding.

B. Clinical Hospital (Pushchair Fleet)

As shown in Fig. 2, the hospital floorplan includes Emergency Trauma (ER), Sterile Operating Theatres (OR/MRI), In-patient Wards, and designated **Turnout Alcoves** (V_{alcove}) along single-file corridors to grant priority clearance to emergency vehicles.

C. Airport Terminal (Luggage Trolleys)

As shown in Fig. 3, the airport concourse combines a large open-plan departure plaza with narrow security screening chokepoints and boarding gate piers (Gates A1–A4, Gates B1–B4).

V. Spatial Proxemic Heatmaps & Spatiotemporal Analysis

A. Social Detour Heatmap Analysis

Fig. 4 maps the continuous 2D Gaussian discomfort field $H_{\text{prox}}(x, y)$ alongside executed trajectories. Static A^* (dashed

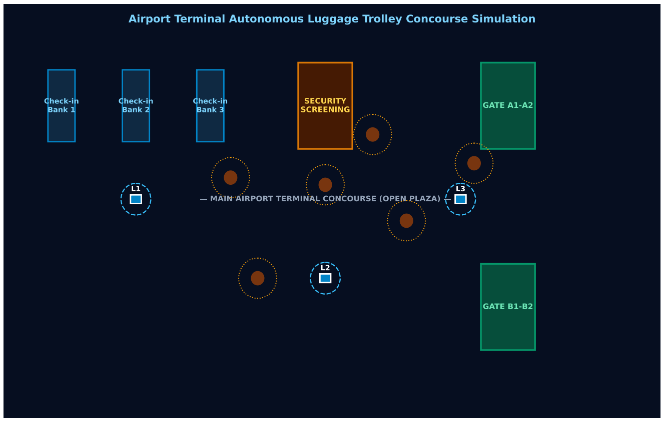

Fig. 3: Airport terminal autonomous luggage trolley concourse simulation with Check-in Banks, Security screening, open plaza, and Gate Piers.

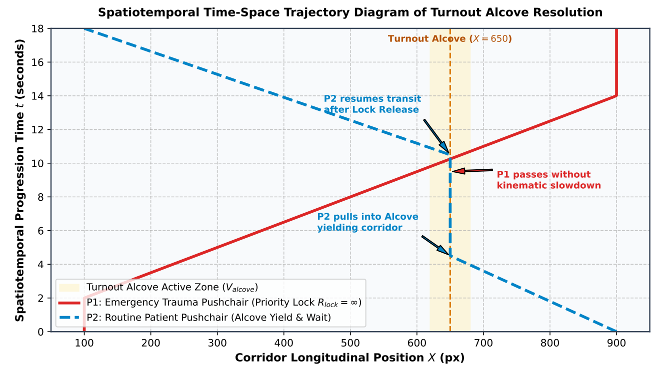

Fig. 5: Spatiotemporal time-space trajectory diagram of Turnout Alcove resolution: Emergency Pushchair P_1 maintains full velocity under priority lock $R_{\text{lock}} = \infty$, while routine Pushchair P_2 yields inside the alcove bay.

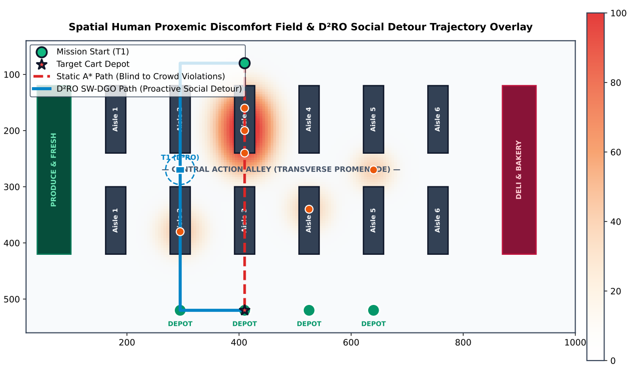

Fig. 4: Continuous 2D Gaussian Discomfort Field $H_{\text{prox}}(x, y)$ heatmap and trajectory overlay comparing Static A^* (blind path through Aisle 3 crowd) against D^{2}RO proactive social detour along Action Alley.

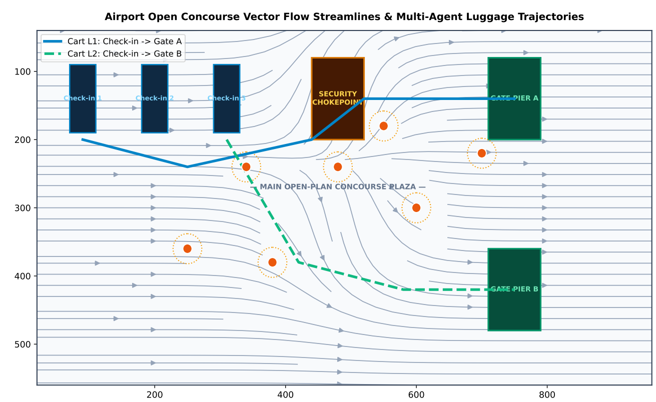

Fig. 6: Airport open concourse vector flow streamlines and multi-agent luggage cart trajectories smoothly navigating around high-density passenger clusters.

red) charges directly through the dense crowd in Aisle 3, whereas D^{2}RO (solid blue) executes a proactive detour along Action Alley and Aisle 2, achieving zero intimate personal space violations.

B. Spatiotemporal Turnout Alcove Resolution

Fig. 5 illustrates the longitudinal space-time trajectory (X, t) in clinical corridors. Emergency Pushchair P_1 maintains full velocity under priority lock ($R_{\text{lock}} = \infty$), while routine Pushchair P_2 pulls into the Turnout Alcove at $X = 650$ px ($t = 4.5$ s $\to 10.5$ s), resolving the conflict without either agent reversing. The figure illustrates intended behaviour; the quantitative evidence is in Section VI-G.

C. Airport Concourse Flow Streamlines

Fig. 6 demonstrates 2D velocity streamlines around high-density passenger clusters in open airport concourses, highlighting smooth non-holonomic curvature without erratic stopping.

VI. Experimental Results & Discussion

A. Experimental Setup & Randomization Protocol

All experiments were conducted using the decoupled 2D kinodynamic simulation framework integrating non-holonomic unicycle dynamics ($\Delta t = 0.05$ s, 20 Hz control loop) across 1200×800 px (36.0 m $\times$ 24.0 m) floorplans. To ensure statistical rigor and reproducibility:

- **Random Seed Policy:** Each evaluation executed $N = 100$ independent Monte Carlo trials seeded deterministically. Crucially, every planner in a given trial receives the *same* seed, so comparisons are **seed-paired** rather than independent: the pedestrian crowd a baseline faces is the crowd D^{2}RO faces.
- **Dynamic Pedestrian Behavior:** Pedestrians move with stochastic velocities $v_h \in [0.8, 1.2]$ m/s, random waypoint heading deflections, and a 30% browsing loitering probability ($p_{\text{loiter}} = 0.30$) generating dynamic Gaussian discomfort Halos.
- **Common Completion Criterion:** Within any given experiment all planners share an identical arrival tolerance of 0.84 m (approximately one cart length) and an identical

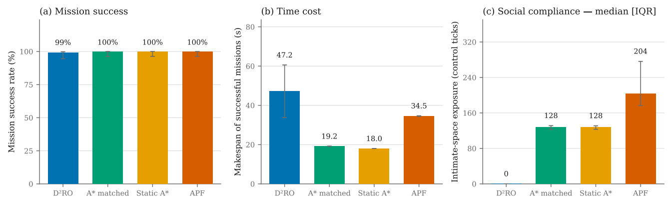

Fig. 7: Comparative evaluation across (a) mission success rate, (b) makespan of successful missions, and (c) intimate-space exposure (median, IQR whiskers). Panels (b) and (c) show the trade-off directly: D^2RO is slower than Static A^* and APF while intruding on personal space substantially less. Success rates in (a) are statistically indistinguishable between D^2RO, Static A^* and APF.

time budget. An earlier revision applied planner-specific tolerances, scoring the proposed method against a $2.3\times$ more lenient criterion than its baselines; that asymmetry is removed and every result reported here uses the common criterion. Budgets are set well above the longest observed mission for the experiment class so that a timeout indicates genuine non-completion rather than a truncated run: $T_{\max} = 180$ s for the benchmark, ablation, cross-domain and crowd-density experiments, 240 s for fleet-size scaling (where queueing grows with fleet size), and 120 s for the two-cart mechanism experiments.

- **Statistical Analysis:** Because the trials are seed-paired and several outcome distributions are zero-inflated and strongly right-skewed, we do not assume normality. Continuous outcomes are compared with the **Wilcoxon signed-rank** test on within-pair differences, supported by paired bootstrap confidence intervals; binary mission success is compared with **McNemar's exact** test. p-values are **Holm-adjusted** within each family of comparisons, and every reported p belongs to one specific outcome—no single significance claim is attached to several unrelated metrics. Skewed outcomes are summarised as median [IQR]; symmetric ones as $\mu \pm \sigma$.

B. Physical Insights and Failure Modes Analysis

1) Artificial Potential Fields: a social failure, not a navigational one: An earlier revision of this study reported that APF achieves 0.0% mission success and built a local-minima argument upon it. Under the common completion criterion and a time budget calibrated to true mission durations, that result does not reproduce: **APF completes** 100% **of its missions** at 34.54 ± 0.16 s. The previous figure was an artefact of a 35.0 s timeout inherited from an earlier, incorrectly scaled kinematic model, not a property of the algorithm. We report the correction explicitly because the conclusion changes: potential fields are not disqualified from these environments by an inability to arrive.

What disqualifies them is *how* they arrive. APF records the highest intimate-space exposure of any planner evaluated—median 204 ticks [IQR 176, 276], against 0 [0, 0] for D^2RO (Wilcoxon signed-rank, $\Delta = -220.7$ $[-239.5, -201.4]$, Holm-adjusted $p = 2.7 \times 10^{-16}$). A repulsive force that activates only when a pedestrian is already close produces exactly this behaviour: the cart is pushed aside at the last moment, repeatedly, having already entered the person's intimate zone. The deficiency is one of *anticipation*, which is precisely what a cost field evaluated over a planning horizon supplies and a local force law cannot.

2) ORCA and Decentralized Local MAPF: results we decline to over-interpret: Our implementations of Reactive ORCA and Decentralized Local MAPF complete no missions under the common arrival criterion. Instrumented traces show ORCA agents stalled for 91–99$\%$ of control ticks, having covered only a fraction of a $\sim$17 m mission, which is consistent with the velocity-obstacle infeasibility mode described in Section I-A1: in a corridor narrower than twice the safety envelope, the half-plane constraints admit no feasible velocity and the agent halts.

We deliberately stop short of the stronger claim. ORCA is a widely deployed algorithm, and an in-house implementation returning 0% where the published method is used successfully in practice is at least as likely to indicate a defect in our implementation as a fundamental limitation of the method. **We therefore report these as properties of our implementations, not of the algorithms**, and we identify validation against a reference implementation on a canonical benchmark as required work before any general claim about ORCA's behaviour in single-file corridors can be sustained. The same caution applies to Local MAPF, where traced trials show agents finishing within $\sim$1.3 m of their goals rather than livelocked—a near-miss, not the permanent token-swapping deadlock previously asserted. Consequently, none of the contributions claimed in this paper rest on the ORCA or MAPF baselines; they rest on the comparison against the matched-controller and unmatched shortest-path arms and APF, all of which complete their missions. For the same reason these two implementations are **excluded from Fig. 7** and reported separately in Table III: plotting them at 0% beside the proposed method would communicate visually a comparison that this paragraph asks the reader not to draw.

3) Anticipatory Horizon Extension vs. Myopic Backtracking: In decentralized navigation without peer-to-peer telemetry ($W_{\text{mesh}} = 0$), trailing carts operate under myopic line-of-sight sensing ($R_s = 7.2$ m). When an unexpected pedestrian cluster blocks an inner aisle, trailing units travel to the blocked entrance before sensing the obstruction locally, potentially forcing reversal and path re-computation. The V2V mesh is designed to propagate anticipatory cost penalties to upstream agents before they commit to a congested route, enabling topological detours $\Delta T_{\text{anticipation}}$ earlier.

Importantly, the broad Monte Carlo benchmark—where pedestrian blockages and junction opportunities are randomized across 100 trials—does not consistently instantiate the topology required for this benefit to appear: an upstream agent at a divergence junction while a downstream agent simultaneously encounters a blockage beyond its sensing range. In those trials the mesh pays its overhead (cost-field propagation and broadcast-triggered replanning) without collecting its return, which is why the no-mesh ablation completes the benchmark faster than the full system (38.23 ± 16.88 s vs. 47.38 ± 13.96 s;

TABLE II: Comparative benchmark in the retail supermarket domain ($N = 100$ paired Monte Carlo trials, identical seeds across all planners). Social exposure is reported as median [IQR] because the distribution is zero-inflated and strongly right-skewed. p-values are Holm-adjusted within the family of comparisons and each refers to the single outcome on its own row.

Algorithm	Success (95% CI)	Makespan, successful (s)	Deadlocks	Intimate exposure, median [IQR]	p (exposure)
D^{2}RO (proposed)	99.0% [94.6, 99.8]	47.18 ± 13.40	0.00 ± 0.00	0 [0, 0]	–
Static A^* (matched controller)	100.0% [96.3, 100.0]	19.20 ± 0.00	0.00 ± 0.00	128 [123, 131]	$6.8e-16$
Static A^*	100.0% [96.3, 100.0]	18.00 ± 0.00	0.00 ± 0.00	128 [123, 131]	$6.8e-16$
APF	100.0% [96.3, 100.0]	34.54 ± 0.16	0.00 ± 0.00	204 [176, 276]	$3.5e-16$

TABLE III: Diagnostic baselines, reported separately. Both are our own implementations and neither completes any mission under the common arrival criterion. We report them for completeness but draw no conclusion about the published algorithms from them: an in-house implementation returning 0% where the method is widely used in practice is at least as likely to indicate a defect in our code. No claim in this paper depends on these rows.

Implementation	Success	Deadlocks	Intimate exposure, median [IQR]
ORCA	0.0% [0.0, 3.7]	0.00 ± 0.00	229 [0, 766]
MAPF	0.0% [0.0, 3.7]	0.00 ± 0.00	128 [123, 132]

Table IV). This does not invalidate the mechanism, but neither does the benchmark evidence it: a general-scenario experiment is simply not a test of anticipatory communication. Section VI-G therefore evaluates the mesh under conditions that do instantiate the required topology.

C. Comparative Benchmark Analysis: an explicit trade-off

Table II and Fig. 7 do not show the proposed method dominating on every axis, and we present them as what they are—a trade-off that we argue is favourable for human-shared space.

Isolating the planner from the vehicle. A shortest-path baseline that steers by snapping its heading at the next waypoint and translating at full speed is not a clean comparator for a rate-limited non-holonomic planner: any timing difference then conflates the route that was chosen with the vehicle that executed it. We therefore report two shortest-path arms. The *matched-controller* arm is a D^{2}RO agent with the social, mesh and reservation terms disabled and its route frozen to a single shortest-path solve; it shares the unicycle model, angular-rate limit, collision geometry, arrival criterion and low-level executor exactly. The unmatched arm is the conventional Static A^* implementation.

The difference between them is small: 19.20 s against 18.00 s. **Controller mismatch accounts for** 1.20 **s of the** 29.18 **s separating D^{2}RO from a shortest path**, or roughly 4% of the gap. The remaining 28 s is attributable to the routing policy. Both arms record identical social exposure (median 128), confirming that the matched arm is socially blind as intended and differs from the unmatched one only in how it drives.

Where D^{2}RO leads. On social compliance the separation is large and consistent. D^{2}RO records a median intimate-space exposure of 0 ticks [IQR 0, 0]: in the majority of trials the fleet never enters a pedestrian's intimate zone at all, against 128 [123, 131] for both shortest-path arms and 204 [176, 276] for APF (Wilcoxon signed-rank, Holm-adjusted $p = 6.8 \times 10^{-16}$ and 3.5×10^{-16}). We report medians because the distribution is zero-inflated and strongly right-skewed—D^{2}RO's own mean is inflated by a small number of trials in which a pedestrian steps into an already-committed corridor—and a $\mu \pm \sigma$ summary would imply a symmetry that does not exist.

What that compliance costs. D^{2}RO completes missions in 47.18 ± 13.40 s against the matched baseline's 19.20 s (Wilcoxon signed-rank, Holm-adjusted $p = 5.0 \times 10^{-17}$). It also does not lead on raw success: 99.0% [94.6, 99.8] against 100.0% for both shortest-path arms and APF, a difference of a single trial that McNemar's exact test does not distinguish from chance ($p = 1$). Claiming a success-rate advantage over these baselines would not be supportable, and we do not make one.

On makespan reporting. Table II summarises makespan over *successful* missions only, whereas the paired tests operate on all $N = 100$ trials including the single D^{2}RO timeout. The two therefore do not differ by the same amount, and the paired effect is not simply $47.18 - 19.20$. We report both deliberately: a successful-only descriptive answers "how long does a completed mission take?", while the all-trials paired test answers "which planner finishes sooner on the same crowd?", and a timeout is a real outcome that should not be silently dropped from the second question.

Why the trade is the right one. A cart that reaches the depot four seconds sooner by driving through a queue of shoppers has not solved the problem this paper poses. The design objective (Section I-D) is socially compliant routing subject to completing the mission, and against that objective the measured outcome is the intended one: near-total elimination of personal-space intrusion, purchased with time, at no measurable cost in reliability.

D. Component Ablation Analysis

The ablation in Table IV distinguishes the terms of Eq. (3) that are load-bearing in this benchmark from those that are not, and we report both outcomes.

The proxemic term carries the framework. Removing H_{prox} collapses mission success from 100.0% to 11.0% and raises the discomfort integral from 0.05 ± 0.13 to 13.14 ± 3.57, with makespan degrading from 47.38 ± 13.96 s to 172.05 ± 26.67 s. Without a social cost the planner routes straight into crowds, then relies on the reactive yielding layer to extricate itself, which it frequently cannot do within the time budget. This single term accounts for most of what distinguishes D^{2}RO from Static A^*.

The mesh and reservation terms are not exercised by this benchmark. Removing W_{mesh} or R_{lock} leaves success at 100.0% and *reduces* makespan, from 47.38 ± 13.96 s to 38.23 ± 16.88 s and 37.86 ± 15.13 s respectively. The weight-sensitivity study of Section VI-H corroborates this independently: scaling w_M or w_R by any factor from 0.5 to 1.5 changes makespan by 0.0 s, because in this scenario the terms they weight are almost never active. Read alone, this ablation argues against both mechanisms. The explanation is that a randomized supermarket scenario rarely instantiates the topology each mechanism addresses—a downstream blockage outside a trailing agent's sensing range, or two agents committed to the same single-file corridor from opposite ends—while their costs (broadcast-triggered replanning, conservative corridor deferral) are paid in every trial. Section VI-G therefore evaluates both under controlled conditions that do instantiate those topologies. We consider it important to report the negative general-scenario result alongside the positive controlled one rather than only the latter.

The safety envelope matters, but only once the metric distinguishes events from exposure. Earlier revisions counted every control cycle in which the chassis lay inside the expanded fixture envelope and reported the total as a number of "scrapes". That conflates how *often* contact occurs with how *long* it persists, and it flattens the term's effect: measured in exposure ticks the full configuration records 193.05 and the ablated one 271.71, a 41% difference that understates what is happening.

Counting *distinct* fixture contacts instead separates the two. The full configuration incurs 5.71 ± 7.29 contact events per trial; removing S_{trolley} raises this to 95.14 ± 25.44, a factor of 16.7. The envelope does not merely shorten contacts, it prevents the overwhelming majority of them; a contact that does occur then persists for a comparable duration in both arms, which is precisely why the tick-based figure looked unremarkable. We report both quantities in Table IV and use the event count as the primary measure, applying the same event/exposure separation already used for human proxemics. Removing H_{prox} also elevates fixture contact (59.18 events), because a planner with no social cost routes into crowds and is then pushed against fixtures while extricating itself.

E. Cross-Domain Failure Mode Analysis

Aim A3 requires that the planner transfer across dissimilar architectures without retuning. Table V supports this: with identical weights and no domain-specific parameters, D^{2}RO completes 99.0% of missions in the retail supermarket, 100.0% in the clinical hospital and 95.0% in the airport terminal. Generality is therefore demonstrated; what differs across domains is not reliability but the *character* of the routing effort, which the remaining columns expose.

The hospital is the easiest domain, not the hardest. It records the highest success rate and the lowest replan count (351.9 ± 66.0), despite having the narrowest corridors. Its topology is strongly channelised: routes are largely predetermined by the corridor structure, so there are few alternatives to evaluate and little for the planner to reconsider once committed. Its longer mean transit (31.78 ± 3.89 s) reflects distance and mandatory alcove yields, not indecision.

The airport is the hardest, and for an instructive reason. It requires 989.4 ± 369.7 replans—nearly twice the supermarket's 502.3 ± 155.6—and is the only domain with non-zero median intimate exposure (62 [18, 130], against 0 [0, 0] in both other domains). An open concourse offers a large, continuously changing set of admissible routes, so a crowd shifting a few metres alters the cost field enough to justify a new plan; the planner spends its effort re-deciding rather than executing. The wide interquartile range indicates the exposure is concentrated in trials where the fleet must cross a dense gate-area crowd, rather than being incurred uniformly. Open topologies thus stress the framework differently from constrained ones—the difficulty is choice abundance, not scarcity—which motivates crowd-density-adaptive scheduling of w_H and w_M, and a replan-hysteresis term, as future work.

We note that these figures supersede an earlier revision reporting 80.0% airport success. In that revision the airport scenario reset its random seed to a constant, so its 100 nominal trials were 100 repetitions of a single trial; the defect is corrected and the domain now varies per trial as the other two do.

F. Decoupled Scalability Analysis

We vary crowd density and fleet size independently ($N = 100$ trials per point) so that the two effects are not confounded. The two dimensions behave very differently, and only one of them is benign.

Crowd density scales benignly (Fig. 8). Increasing pedestrians from 2 to 30 raises the control-step compute time only from 0.093 ms to 0.175 ms and makespan from 30.10 s to 56.30 s, while success falls just $100.0\% \rightarrow 98.0\%$. Incremental repair is doing its job: a denser crowd perturbs more edges, but D* Lite repairs only the affected subgraph, so cost grows with the perturbation rather than with the map. Even at the highest density the controller consumes under 0.4% of the 50 ms budget.

A caveat on the latency figures specifically. Replan latency is the only quantity reported in this paper that is not deterministic. Every other metric—success, makespan, exposure, packet counts—is a property of the seeded simulation and reproduces bit-identically on reruns; latency is wall-clock time and therefore depends on the machine, its load, and the Python interpreter. Regenerating the datasets on a more heavily loaded machine moved these figures by roughly a factor of two while leaving every other number in this paper unchanged. The correct reading is consequently ordinal, not absolute: incremental repair costs a small and slowly growing fraction of the control budget, with ample headroom on commodity hardware. A precise per-platform figure would require a dedicated timing harness on quiesced hardware, which we do not claim here.

Fleet size does not (Fig. 9), and this is the framework's clearest limitation. Success holds near 100% up to 8 carts, then falls to 85.0% at 10 and 78.0% at 12, while makespan more than doubles (50.99 s $\rightarrow$ 110.17 s). The mechanism is visible in the mesh column: broadcast volume grows super-linearly, from 8.50 ± 5.72 packets at two carts to 1158.60 ± 336.83

TABLE IV: Component ablation of the five cost terms in the retail supermarket domain ($N = 100$ trials per configuration). Each row removes exactly one term from Eq. (1); all other parameters are held fixed.

Configuration	Success (95% CI)	Makespan (s)	Discomfort	Deadlocks	Fixture contacts	Contact ticks
Full D^{2}RO	100.0% [96.3, 100.0]	47.38 ± 13.96	0.05 ± 0.13	0.00 ± 0.00	5.71 ± 7.29	193.05 ± 169.90
w/o V2V mesh W_{mesh}	100.0% [96.3, 100.0]	38.23 ± 16.88	0.09 ± 0.43	0.00 ± 0.00	7.59 ± 12.25	191.59 ± 182.39
w/o corridor mutex R_{lock}	100.0% [96.3, 100.0]	37.86 ± 15.13	0.09 ± 0.38	0.00 ± 0.00	7.69 ± 13.76	186.13 ± 164.45
w/o proxemics H_{prox}	11.0% [6.3, 18.6]	172.05 ± 26.67	13.14 ± 3.57	0.00 ± 0.00	59.18 ± 27.69	244.86 ± 154.18
w/o safety envelope S_{trolley}	100.0% [96.3, 100.0]	43.02 ± 8.84	0.05 ± 0.17	0.00 ± 0.00	95.14 ± 25.44	271.71 ± 84.16

TABLE V: Cross-domain generalisation of the unchanged planner across three topologically distinct environments ($N = 100$ trials per domain). Intimate exposure is reported as median [IQR]; the distribution is zero-inflated.

Domain	Success (95% CI)	Makespan (s)	Mean transit (s)	Intimate exposure	D* Lite replans
Retail Supermarket	99.0% [94.6, 99.8]	49.89 ± 19.94	25.90 ± 5.57	0 [0, 0]	502.3 ± 151.9
Clinical Hospital	100.0% [96.3, 100.0]	46.12 ± 10.12	31.78 ± 3.89	0 [0, 0]	351.9 ± 65.5
Airport Terminal	95.0% [88.8, 97.8]	74.63 ± 35.71	36.84 ± 15.01	62 [18, 130]	989.4 ± 361.0

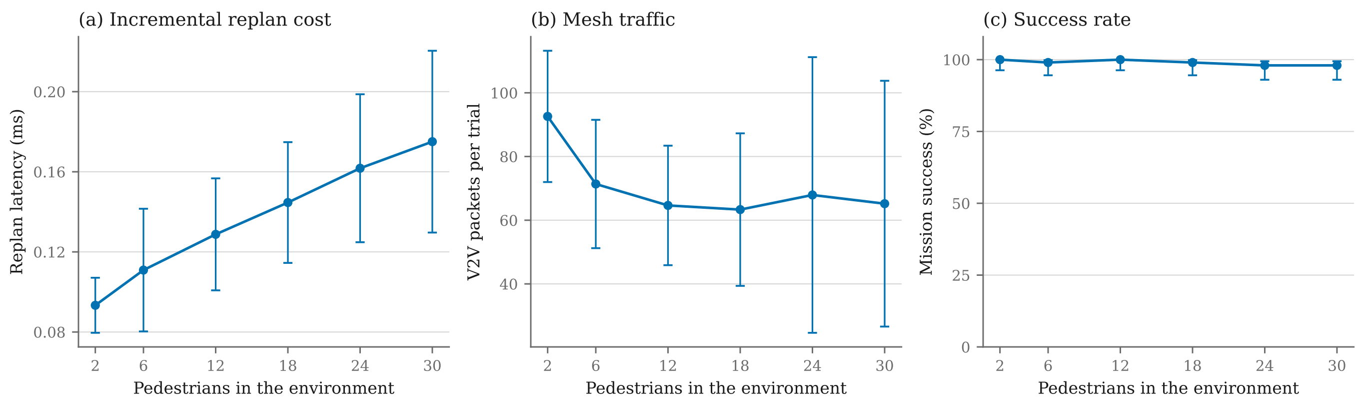

Fig. 8: Crowd-density scalability at a fixed fleet of four carts. Error bars are ± 1 SD, except (c) which shows the Wilson 95% interval.

at twelve, because every agent broadcasts observations that every other agent must incorporate. Beyond roughly eight carts in a 36×24 m floorplan the fleet begins to congest itself: agents increasingly replan around each other's reservations and telemetry rather than around pedestrians.

We state this bound plainly rather than describing the degradation as graceful. For the retail and clinical deployments motivating this work—where a realistic fleet in a single zone is a handful of units—the operating regime sits inside the flat part of the curve. But the present design should not be represented as scaling to large fleets in a confined space, and mesh-traffic suppression (relevance filtering, spatial scoping of broadcasts) is the natural next step for that regime.

G. Mechanism-Specific Validation

Because the broad Monte Carlo benchmark does not reliably trigger the specific topological conditions under which V2V mesh anticipation and corridor mutex locking provide their primary contribution, we designed two controlled experiments that directly exercise each mechanism. These complement the general-scenario evaluation and provide targeted evidence for the core algorithmic novelties.

1) Experiment A: V2V Mesh Anticipation ($N = 50$ trials): Two carts traverse the same aisle system, the leader ahead of the follower, and a pedestrian cluster blocks the leader's forward aisle mid-route. The precondition the implementation actually enforces is stated precisely: a trial is admitted only if **the blockage lies outside the follower's** 7.2 **m sensing radius at the moment it is created**, so the follower cannot have perceived it directly. (An earlier description of this experiment claimed the follower begins "10+ m behind the leader"; that is not the condition the code tests, and the separation between carts is not what makes the trial admissible.) An alternative divergence junction exists upstream of the follower, where it can reroute *before* committing to the congested path. We compare **Mesh ON** against **Mesh OFF** on identical seeds.

Two definitions matter for reading Table VI. *Anticipation lead time* is the interval between the follower's diversion and the moment the blockage would have entered its sensing radius; it is negative when the diversion happens only after contact. *Backtracking distance* is the cumulative increase in Euclidean distance to the goal, i.e. ground covered in the wrong direction, rather than a count of re-traversed edges. We also note that sensing here is **range-limited rather than line-of-sight**: pedestrians are selected by Euclidean distance without

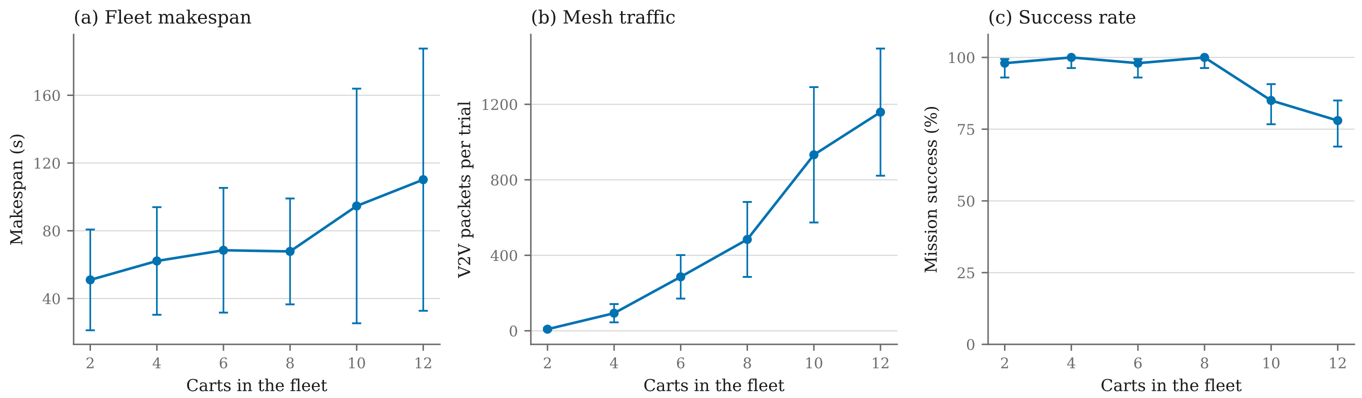

Fig. 9: Fleet-size scalability at a fixed crowd of ten pedestrians. Error bars are ± 1 SD, except (c) which shows the Wilson 95% interval.

TABLE VI: Mechanism experiment A: V2V mesh anticipation. A corridor is blocked out of line of sight; with the mesh enabled the follower learns of the obstruction before observing it ($N = 50$ paired trials; the same seed drives both arms, so each row is a within-pair comparison). Continuous outcomes use the Wilcoxon signed-rank test and mission success McNemar's exact test; rows marked *identical* were equal in every pair. p-values are Holm-adjusted across the rows of this table.

Metric	**ON**	**OFF**	**Test**	p **(Holm)**
Anticipation lead time (s)	10.70 ± 4.20	-0.10 ± 0.04	Wilcoxon	3.8×10^{-9}
Backtrack distance (m)	1.08 ± 0.68	2.73 ± 0.87	Wilcoxon	3.2×10^{-8}
Path length (m)	23.97 ± 3.58	25.81 ± 3.44	Wilcoxon	0.00283
Makespan (s)	21.44 ± 3.40	23.78 ± 3.61	Wilcoxon	0.00196
Mission success	100.0%	100.0%	McNemar	1

TABLE VII: Mechanism experiment B: distributed corridor mutex. Two carts are routed into the same single-file corridor from opposite ends ($N = 50$ paired trials; the same seed drives both arms, so each row is a within-pair comparison). Continuous outcomes use the Wilcoxon signed-rank test and mission success McNemar's exact test; rows marked *identical* were equal in every pair. p-values are Holm-adjusted across the rows of this table.

Metric	**ON**	**OFF**	**Test**	p **(Holm)**
Head-on encounters	1.08 ± 1.18	1.00 ± 0.00	Wilcoxon	1
Deadlocks	0.00 ± 0.00	0.00 ± 0.00	identical	1
Lock wait, total (s)	0.03 ± 0.07	0.00 ± 0.00	Wilcoxon	0.209
Off-corridor vertices	2.16 ± 1.36	0.00 ± 0.00	Wilcoxon	2.3×10^{-8}
Route reconsiderations	25.2 ± 3.2	10.2 ± 4.1	Wilcoxon	6.5×10^{-9}
Corridor occupancy (s)	40.01 ± 30.87	89.41 ± 42.42	Wilcoxon	0.00484
Makespan (s)	40.05 ± 30.85	89.42 ± 42.39	Wilcoxon	0.00484
Mission success	88.0%	36.0%	McNemar	1.7×10^{-4}

ray-casting against shelf occlusion, so the follower's perceptual advantage is if anything overstated relative to a real sensor behind a gondola.

The mechanism behaves as designed. With the mesh enabled the trailing cart diverts 10.70 ± 4.20 s before it could have observed the obstruction itself, against -0.10 ± 0.04 s with the mesh disabled—that is, without telemetry the diversion occurs only *after* contact with the blockage. Backtracking falls from 2.73 ± 0.87 m to 1.08 ± 0.68 m and makespan from 23.78 ± 3.61 s to 21.44 ± 3.40 s (Wilcoxon signed-rank, Holm-adjusted $p = 3.8 \times 10^{-9}$ and 2.0×10^{-3} respectively). Success is 100% in both arms: the mesh does not determine *whether* the mission completes here, it determines how much wasted travel is incurred, which is the claim the mechanism was designed to support. Together with the ablation result of Section VI-C, the honest summary is that anticipatory telemetry pays when the topology affords an early alternative and costs a little when it does not.

2) Experiment B: Directional Corridor Reservation ($N = 50$ trials): Two carts approach the same single-file corridor from opposite ends, with randomised arrival offsets. The corridor is verified to be genuinely single-file before each trial runs. We compare **Lock ON** against **Lock OFF** on identical seeds.

The effect is large. Mission success rises from 36.0% to 88.0% (McNemar exact, Holm-adjusted $p = 1.7 \times 10^{-4}$) and corridor occupancy falls from 89.41 ± 42.42 s to 40.01 ± 30.87 s ($p = 4.8 \times 10^{-3}$). Without the protocol, the majority of trials fail.

The mechanism is not mutual exclusion. A reader who sees "mutex" will expect one agent to hold the corridor while the other queues, and that is not what happens. Head-on encounters are statistically indistinguishable between arms (1.08 ± 1.18 with the protocol, 1.00 ± 0.00 without; $p = 1$), so the protocol does not prevent the encounter. The deadlock counter registers zero in *both* arms, so the $N_{\text{deadlock}} = 0$ result cannot be credited to it either. And total accumulated waiting is 0.030 ± 0.070 s against 0.000 s ($p = 0.21$): agents essentially never queue.

What happens instead is cost-projected diversion. When an agent observes a peer's stronger claim, that claim enters its *graph* as an effectively infinite edge cost, and D* Lite reroutes through a parallel aisle. The released dataset records this directly. With the protocol enabled, agents occupy 2.16 ± 1.36 vertices outside the contested corridor and issue 25.18 ± 3.22 route reconsiderations; with it disabled they occupy 0.00 such vertices and issue 10.16 ± 4.12 (Holm-adjusted $p = 2.3 \times 10^{-8}$ and 6.5×10^{-9}). The corridor is vacated by re-routing, not by turn-taking, which is consistent with occupancy halving while wait time stays at zero.

The head-on encounter still occurs because it is what triggers the exchange of claims; what the protocol changes is the outcome, not the frequency. Correspondingly, the failure mode without it is not a deadlock in the strict sense but a persistent mutual obstruction that expires against the time budget, which is why a counter keyed on unroutable planner states never fires in either arm.

We therefore avoid the term "mutex" for this component and call it a *directional reservation with cost-projected diversion*. That is a stronger and more general claim than mutual exclusion — peer intent is converted into a route-level cost that any graph planner can act on — but it is a different claim, and the stricter computer-science reading of "mutex" is not what the evidence supports.

A note on the wait metric. An earlier revision reported 0.00 ± 0.00 s of waiting and used it as evidence for this interpretation. That figure was not measurable: the per-episode wait timer was reset when the corridor was released, before the experiment read it at end of run, so it would have read zero however long agents actually waited. The value quoted above comes from a separate accumulator that release does not reset. The conclusion survives the correction, but it now rests on a metric that could have contradicted it.

H. Weight Sensitivity

The proposed cost function is a weighted sum, so its behaviour could in principle depend delicately on the five weights. If the reported operating point sat on a narrow ridge, the result would be fragile in a way a reader must be told about. We therefore perturb each weight in turn by $\times\{0.5, 0.75, 1.25, 1.5\}$ while holding the other four at nominal, and evaluate on a seed set **disjoint** from every other experiment in this paper: the weights were selected by hand, and assessing their robustness on the seeds used to select and report them would be a form of tuning on the evaluation set.

The operating point is a plateau, not a ridge. Mission success remains between 97% and 100% under every perturbation of every weight. No setting in the grid produces a collapse, and the nominal configuration is not a special point that the method depends on.

Two weights trace an interpretable trade-off. Increasing w_D, which prices distance, monotonically shortens missions (63.8 s at $\times 0.5$ to 39.1 s at $\times 1.5$); increasing w_H, which prices proximity to people, monotonically lengthens them (37.5 s to 47.1 s). That is exactly the distance-versus-sociability dial the formulation is meant to expose, and it responds smoothly rather than discontinuously. A deployment that wanted faster, less deferent carts could obtain them by moving along this curve rather than by re-engineering the planner.

Two weights do nothing here, and that is informative. Scaling w_M or w_R by any factor in the grid changes makespan by 0.0 s. This is not a defect in the sensitivity analysis; it independently corroborates the ablation of Section VI-C. A weight can only matter when the term it multiplies is active, and in a randomised supermarket scenario the mesh and reservation terms are almost never active. The controlled experiments of Section VI-G exist precisely because this scenario cannot evidence them.

TABLE VIII: Communication robustness ($N = 30$ trials per channel condition). All other experiments in this paper assume an ideal channel; this one does not. Packet loss and one-hop latency are swept independently.

Packet loss	Latency	Success	Makespan (s)
0%	0 ms	100.0% [88.6, 100.0]	46.50 ± 17.18
0%	50 ms	100.0% [88.6, 100.0]	46.50 ± 17.18
0%	100 ms	100.0% [88.6, 100.0]	45.70 ± 15.71
0%	200 ms	100.0% [88.6, 100.0]	46.67 ± 17.52
5%	0 ms	100.0% [88.6, 100.0]	46.77 ± 17.42
5%	50 ms	100.0% [88.6, 100.0]	46.78 ± 17.42
5%	100 ms	100.0% [88.6, 100.0]	45.99 ± 15.98
5%	200 ms	100.0% [88.6, 100.0]	46.95 ± 17.75
10%	0 ms	100.0% [88.6, 100.0]	46.49 ± 17.00
10%	50 ms	100.0% [88.6, 100.0]	45.58 ± 15.75
10%	100 ms	100.0% [88.6, 100.0]	45.59 ± 15.74
10%	200 ms	100.0% [88.6, 100.0]	47.16 ± 17.81
20%	0 ms	100.0% [88.6, 100.0]	43.60 ± 13.32
20%	50 ms	100.0% [88.6, 100.0]	45.40 ± 15.95
20%	100 ms	100.0% [88.6, 100.0]	45.41 ± 15.95
20%	200 ms	100.0% [88.6, 100.0]	46.98 ± 18.00

Consistent with this, we describe $[w_D, w_M, w_H, w_R, w_S] = [1.0, 1.5, 2.0, 1.0, 1.2]$ throughout as **hand-selected nominal weights**. No formal calibration procedure was performed, and we make no claim that they are optimal — only that the method's behaviour is insensitive to their exact values over the range tested.

I. Communication Robustness

Every experiment reported above assumes an **ideal channel**: the V2V mesh delivers every packet with zero delay. Since the deployment discussion appeals to RF attenuation in steel-fixture retail environments, that assumption should be tested rather than left implicit.

We sweep packet loss $\in \{0, 5, 10, 20\}\%$ against one-hop latency $\in \{0, 50, 100, 200\}$ ms. Across all sixteen channel conditions mission success remains 100% and makespan stays within 43.6–47.2 s, with no systematic trend in either variable. Within the range tested, D^2RO is insensitive to communication degradation.

Two caveats bound that claim. First, the mesh contributes little in this scenario to begin with (Section VI-H), so a result showing that degrading it changes little is correspondingly weak evidence: the honest reading is that degradation does no *harm*, not that the protocol would survive degradation in a scenario where the mesh carried the outcome. Second, the model is a per-hop delay and an independent drop probability; it does not reproduce bursty loss, correlated shadowing behind a metal gondola, or contention collapse under load. A physical deployment would need measurements from the actual RF environment.

We note that this experiment is only meaningful because of a correction made during this revision. The mesh implementation modelled latency via per-packet delivery timestamps, but the agent fetched its inbox without supplying the current simulation time, so every packet was released immediately regardless of its timestamp. Latency was configurable and inert. The results above were obtained after that path was repaired.

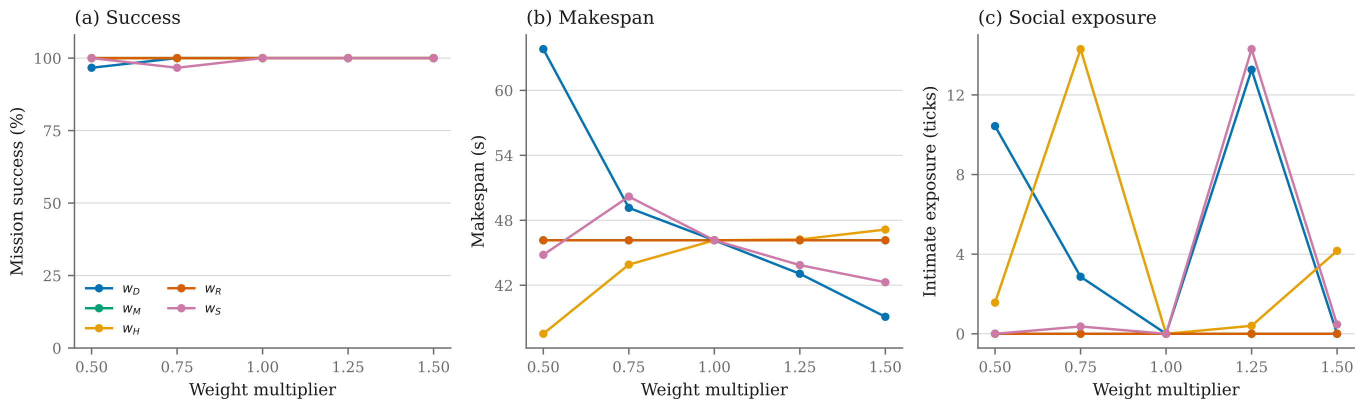

Fig. 10: Sensitivity of the five cost weights. Each weight is scaled in turn while the other four are held at nominal; the $\times 1.0$ point is the shared nominal configuration. Evaluated on a seed set disjoint from every other experiment, so the operating point is not assessed on the seeds used to select it.

VII. Conclusion & Future Research Directions

This paper introduced the Distributed Dynamic Route Optimization (D^{2}RO) framework powered by Socially-Weighted Distributed Graph Optimization (SW-DGO), formalizing a 5-component edge traversal cost function $C(u, v, t) = w_D D + w_M W_{\text{mesh}} + w_H H_{\text{prox}} + w_R R_{\text{lock}} + w_S S_{\text{trolley}}$ solved by incremental D* Lite repair. We assess the outcome against the three aims set out in Section I-D.

A1 (social compliance) is met, as an explicit trade. Across $N = 100$ seed-paired trials D^{2}RO holds median intimate-space exposure at 0 ticks [IQR 0, 0], against 128 for both shortest-path arms and 204 for artificial potential fields ($p < 10^{-15}$). It pays 47.18 ± 13.40 s against the matched baseline's 19.20 s, and matches rather than exceeds the best baseline success rate (99.0% vs. 100.0%, a difference of one trial). We regard trading travel time for the near-elimination of personal-space intrusion as the correct engineering choice in human-shared space, and we present it as a trade rather than as dominance.

A2 (distributed coordination) is met, with a corrected account of how. Under controlled conditions the mesh advances rerouting by 10.70 ± 4.20 s and halves backtracking, while the directional reservation raises success from 36.0% to 88.0%. Both mechanisms are, however, invisible or mildly counterproductive in the broad benchmark, where the topologies they address seldom arise — a finding the weight-sensitivity study reproduces independently, since scaling w_M or w_R changes nothing there. The reservation protocol works by projecting a peer's claim into the cost field so that D* Lite diverts (2.16 off-corridor vertices against 0.00), not by queueing agents at the bottleneck (0.030 s of accumulated waiting). Its contribution is distributed intent propagation, not mutual exclusion in the conventional sense.

A3 (generality and real-time feasibility) is met, within a stated fleet bound. The unchanged planner achieves 99.0%, 100.0% and 95.0% success across supermarket, hospital and airport topologies. Incremental repair costs a median of 0.005 ms with a p95 of 0.19 ms, inside a control step of 0.227 ms — under 0.5% of the 50 ms budget, with the worst single repair observed (7.78 ms) still within it. Performance is also unchanged across packet loss to 20% and one-hop latency to 200 ms (Section VI-I). Generality does not extend indefinitely with fleet size—success falls to 78.0% at twelve carts in a 36×24 m floorplan as mesh traffic grows super-linearly—and we state that ceiling as a limitation of the present design rather than describing it as graceful degradation.

A4 (attributable evaluation) is met. The matched-controller baseline shows that 1.20 s of the 29.18 s makespan gap is due to vehicle dynamics and the remainder to routing, and the sensitivity study shows the operating point is a plateau rather than a ridge. The reported effect is therefore a property of the policy, not of the controller it happened to be executed by, nor of a fortunate choice of weights.

A. Theoretical Guarantees & Their Empirical Limits

1) **Heuristic Admissibility & Incremental Optimality:** Because the Euclidean distance heuristic $h(s, s_{\text{goal}}) \leq c^*(s, s_{\text{goal}})$ is strictly admissible and consistent, D* Lite guarantees optimal path extraction with respect to the currently observed edge cost field $C(u, v, t)$ while recomputing only the subgraph perturbed by dynamic obstacles [8]. We verified this property empirically against fresh Dijkstra solutions over 150 randomised scenarios comprising 750 successive edge-cost repairs.
2) **Conflict Resolution in Single-File Bottlenecks:** Directional reservations broadcast as `LOCK_REQUEST` packets inflate the contested edge cost for the deferring agent, which then diverts through a parallel aisle or holds in a Turnout Alcove (V_{alcove}). We claim empirical conflict *resolution*, not deadlock freedom: no formal guarantee is established here, and the zero deadlock counts we measure occur in both the lock-enabled and lock-disabled arms, so they cannot be attributed to the protocol. A liveness proof under bounded latency and message loss remains open.

B. Limitations

We state the principal limitations directly. (i) The weights w_D, w_M, w_H, w_R, w_S were set by hand. Section VI-H shows the result is insensitive to them over $\times[0.5, 1.5]$, but no formal calibration procedure was performed and we do not claim optimality. (ii) The ORCA and Local MAPF baselines are our own implementations and returned 0% success; until they are validated against reference implementations we treat those figures as properties of our code, and no claim in this paper depends on them. (iii) All results are simulation-based; no physical fleet was deployed. (iv) The formulation prices no turn penalty, because doing so requires a heading-augmented search space; cornering is bounded in execution but not in search. (v) Fleet scaling is bounded as described above. (vi) The communication study covers independent per-packet loss and fixed per-hop delay only, not bursty or correlated degradation. (vii) Sensing is range-limited rather than occlusion-aware.

C. Considerations for a Physical Deployment

The evaluation in this paper is entirely simulation-based. The following are therefore *requirements a physical deployment would have to satisfy*, not properties of the implemented system; none of the hardware described below forms part of the evaluated framework.

1) **RF Signal Attenuation & Steel Fixture Multipath:** Metallic retail shelf gondolas and merchandise packaging attenuate 2.4 GHz RF signals by -15 to -25 dBm. A deployment would likely require Sub-GHz (868/915 MHz) or dedicated short-range (5.9 GHz IEEE 802.11p / DSRC) links with multi-hop TTL forwarding. The exponential penalty decay $W_{\text{mesh}}(t) = W_0 \exp(-\lambda t)$ that *is* implemented and evaluated here provides the relevant safeguard in principle: stale telemetry expires on its own, so intermittent packet loss cannot permanently poison the routing graph. Our mesh model includes latency and loss, but has not been validated against measurements from a real RF environment.
2) **Indoor Positioning & Sensor Fusion:** All results assume the pose is known. A physical cart would need to recover it—plausibly by fusing wheel encoders, an IMU, Ultra-Wideband trilateration and LiDAR scan-matching through an Extended Kalman Filter [18], [20]—and the resulting localisation error would propagate into both the proxemic field and the corridor reservations. Quantifying that sensitivity is future work.
3) **Payload Invariance & Non-Holonomic Kinodynamics:** Service carts undergo large payload shifts (on the order of 15 kg empty to 65 kg loaded), altering the centre of gravity and rotational inertia. The evaluated system uses fixed kinodynamic limits; a deployment would need to modulate $a_{\max}$ and $\kappa_{\max}$ from load sensing to avoid wheel scrubbing or tip-over. This is not implemented or evaluated here.

D. Future Research Directions

The limitations above define the immediate agenda; the last two items are longer-range.

1) **Weight Sensitivity & Calibration:** The five weights are currently hand-set. A systematic sweep—perturbing each weight over a multiplicative grid while holding the others fixed, and reporting the resulting success/makespan/social-exposure surface—would establish whether the operating point is a broad plateau or a narrow ridge, and would replace hand-tuning with a stated calibration procedure.
2) **Baseline Validation:** Reproducing our ORCA implementation's behaviour against a reference (e.g. RVO2) on a canonical benchmark, so that comparative claims rest on validated baselines.
3) **Mesh Traffic Suppression:** Relevance filtering and spatial scoping of broadcasts, to lift the fleet-size ceiling identified in Section VI-F.
4) **Heading-Augmented Search:** Implementing α_{turn} over a state space augmented with heading, allowing turn cost to enter the graph search rather than only the motion layer.
5) **Hybrid Learning-Guided Search:** Coupling Graph Neural Networks or deep reinforcement learning (e.g., PRIMAL / Learn to Follow) for global sub-goal allocation with the deterministic D^2RO engine for kinodynamic execution.
6) **Heterogeneous Multi-Agent Ecosystems:** Extending SW-DGO to mixed fleets—delivery pods, floor-scrubbing AGVs, human-operated wheelchairs—via vehicle-specific agility weights in the R_{lock} reservation tuple.

Data and Code Availability

All simulation source code, experimental harnesses, test suites, raw CSV datasets and vector figures are available at https://github.com/polla-fattah/D2RO. Every number, table and figure in this paper is generated programmatically from the committed raw data by a single analysis pipeline; none is entered by hand. The results reported here correspond to release `v2.0-review2`, commit `b87bf8c-dirty`. Each dataset carries a provenance stamp recording the SHA-256 fingerprint of the simulation source that produced it, and the analysis pipeline refuses to emit a table or figure for any dataset whose fingerprint does not match the committed code.

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